

Sinkhorn Algorithm

1. $\mathbf{K} \leftarrow \exp\{-\mathbf{C}/\varepsilon\}$

2. While not converged:

$$\mathbf{u} \leftarrow \mathbf{a} \oslash \mathbf{K} \mathbf{v}$$

$$\mathbf{v} \leftarrow \mathbf{b} \oslash \mathbf{K}^T \mathbf{u}$$

From EOT Dual to Log-Domain

- Direct way to optimize dual: block coordinate ascent on $H_\varepsilon(\mathbf{f}, \mathbf{g}) \triangleq \langle \mathbf{f}, \mathbf{a} \rangle + \langle \mathbf{g}, \mathbf{b} \rangle - \varepsilon \langle e^{\mathbf{f}/\varepsilon}, \mathbf{K} e^{\mathbf{g}/\varepsilon} \rangle$

- Gradients:

$$\nabla_{\mathbf{f}} H(\mathbf{f}, \mathbf{g}) = \mathbf{a} - e^{\mathbf{f}/\varepsilon} \odot (\mathbf{K} e^{\mathbf{g}/\varepsilon})$$

$$\nabla_{\mathbf{g}} H(\mathbf{f}, \mathbf{g}) = \mathbf{b} - e^{\mathbf{g}/\varepsilon} \odot (\mathbf{K}^\top e^{\mathbf{f}/\varepsilon})$$

- Set gradients to 0, solve for f,g in alternation leads to:

$$\mathbf{f}^{(\ell+1)} = \varepsilon \log \mathbf{a} - \varepsilon \log (\mathbf{K} e^{\mathbf{g}^{(\ell)}/\varepsilon})$$

$$\mathbf{g}^{(\ell+1)} = \varepsilon \log \mathbf{b} - \varepsilon \log (\mathbf{K}^\top e^{\mathbf{f}^{(\ell+1)}/\varepsilon})$$

- This is exactly equivalent to Sinkhorn! Indeed, note that:

$$\mathbf{f}^{(\ell+1)}/\varepsilon = \log(\mathbf{a} \oslash \mathbf{K} e^{\mathbf{g}^{(\ell)}/\varepsilon}) \implies e^{\mathbf{f}^{(\ell+1)}/\varepsilon} = \mathbf{a} \oslash \mathbf{K} e^{\mathbf{g}^{(\ell)}/\varepsilon} \implies \mathbf{u}^{(\ell+1)} = \mathbf{a} \oslash \mathbf{K} \mathbf{v}^{(\ell)}$$

$$\mathbf{g}^{(\ell+1)}/\varepsilon = \log(\mathbf{b} \oslash \mathbf{K}^\top e^{\mathbf{f}^{(\ell+1)}/\varepsilon}) \implies e^{\mathbf{g}^{(\ell+1)}/\varepsilon} = \mathbf{b} \oslash \mathbf{K}^\top e^{\mathbf{f}^{(\ell+1)}/\varepsilon} \implies \mathbf{v}^{(\ell+1)} = \mathbf{b} \oslash \mathbf{K}^\top \mathbf{u}^{(\ell+1)}$$

- Where we used the relation $(\mathbf{u}, \mathbf{v}) = (e^{\mathbf{f}/\varepsilon}, e^{\mathbf{g}/\varepsilon})$,

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Log-Domain Sinkhorn